

Product & Pricing Analysis

Dataset: Brazilian e-commerce marketplace, ~100k orders, Sept 2016 – Aug 2018.

Two questions:

1. **Beyond delivery, what else moves the score?** (price, freight ratio, product weight/size, photo count, description length, number of items)
2. **Which product categories carry high revenue, low revenue, and poor satisfaction?**

1. Imports & Defaults

ready

2. Load Data

```
master_orders    99,441 rows  45 cols
products         32,951 rows  11 cols
order_items      112,650 rows   8 cols
```

```
master columns: ['order_id', 'customer_id', 'order_status', 'order_purchase_timestamp', 'order_approved_at', 'order_delivered_carrier_date', 'order_delivered_customer_date', 'order_estimated_delivery_date', 'delay_days', 'delivery_days', 'handling_days', 'transit_days', 'promise_buffer_days', 'is_late', 'flag_approved_after_carrier', 'flag_delivered_before_carrier', 'review_score', 'is_negative', 'review_creation_date', 'n_items', 'n_sellers', 'total_price', 'total_freight', 'primary_seller_id', 'all_seller_ids', 'n_distinct_categories', 'primary_category', 'freight_ratio', 'is_multi_seller', 'n_payment_methods', 'dominant_payment_type', 'max_installments', 'total_payment_value', 'customer_unique_id', 'customer_state', 'customer_city', 'customer_lat', 'customer_lng', 'seller_state', 'seller_city', 'seller_lat', 'seller_lng', 'delay_bucket', 'geo_distance', 'is_interstate']
```

3. Build the Analytical Base Table

Delivered orders with a review: 95,832

Base with product features: 95,832 orders

Missing product join: 0 rows (0.0%)

	review_score	total_price	freight_ratio	product_weight_g	product_volume_cm3
product_photos_qty	product_description_length	n_items			
count	95832.00	95832.00	95832.00	95832.00	95832.00
95832.00		95832.00	95832.00		
mean	4.16	136.80	0.31	2107.22	15200.26
2.23		791.69	1.14		
std	1.28	207.78	0.31	3762.35	23323.22
1.74		649.88	0.53		
min	1.00	0.85	0.00	0.00	168.00
1.00		4.00	1.00		
25%	4.00	45.90	0.13	300.00	2816.00
1.00		353.00	1.00		
50%	5.00	86.25	0.22	700.00	6400.00
1.00		601.00	1.00		
75%	5.00	149.90	0.38	1813.00	18375.00
3.00		990.00	1.00		
max	5.00	13440.00	21.45	40425.00	296208.00
20.00		3992.00	21.00		

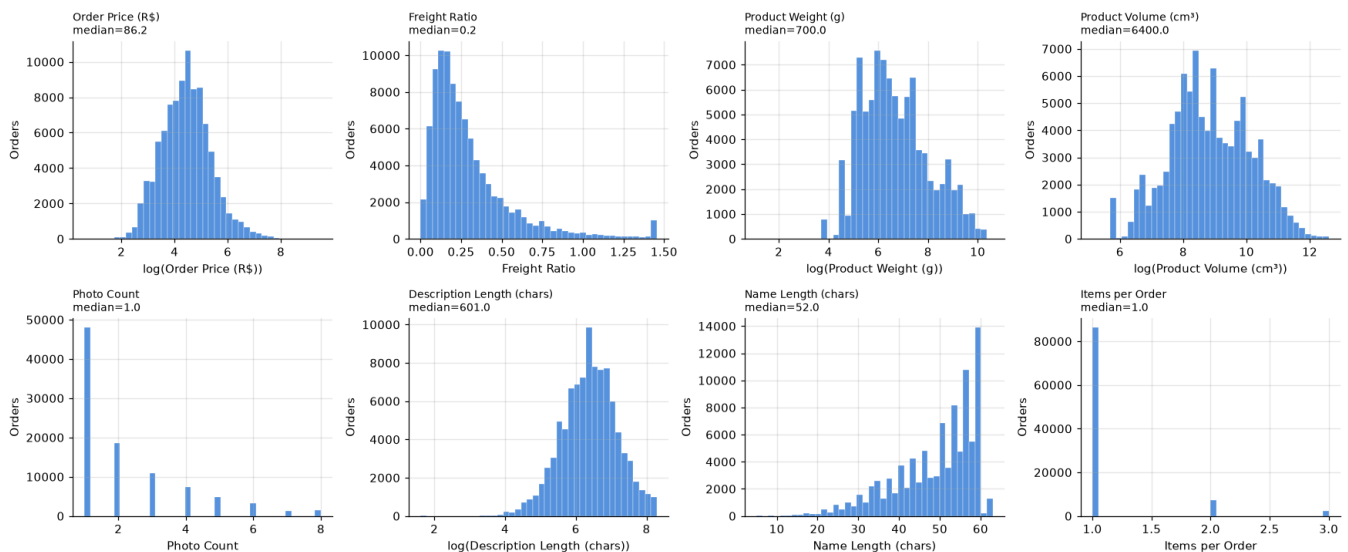
4. EDA on Key Variables

Platform bad-review rate (1-2 stars): 12.76%

Total orders in base : 95,832

Total bad reviews : 12,225

Distribution of product & pricing features



Question 1 — Beyond Delivery, What Else Moves the Score?

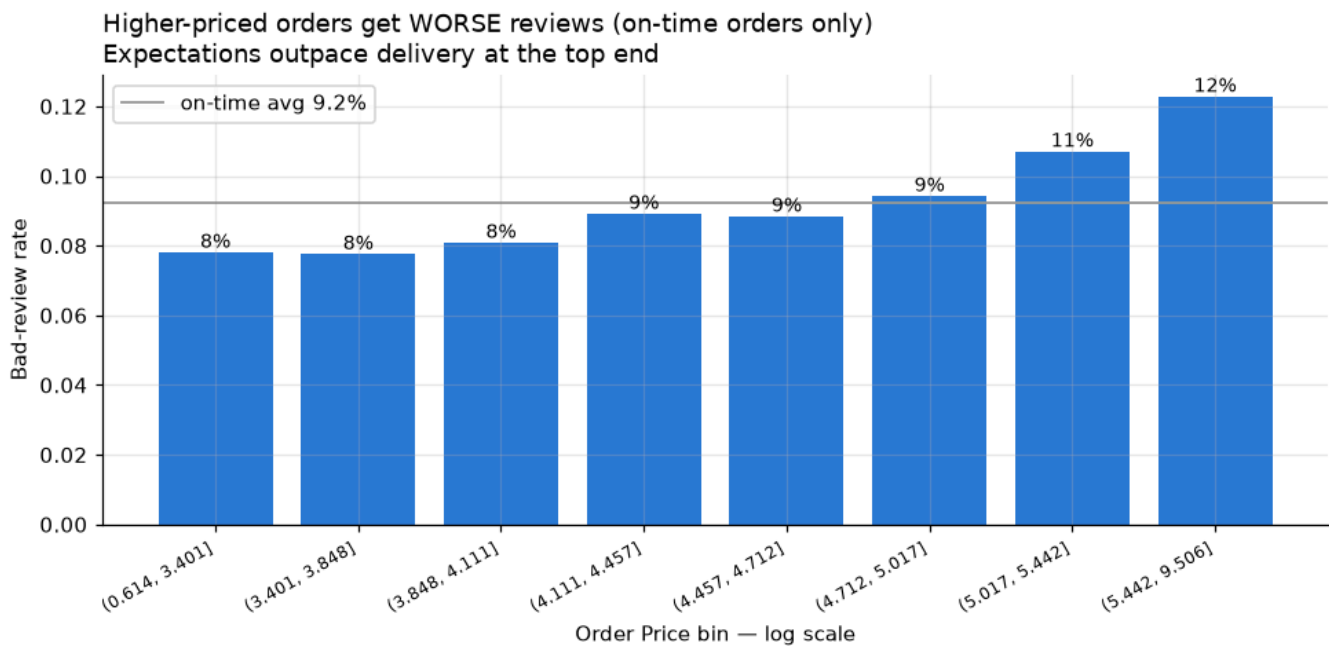
We isolate **on-time orders** to remove the dominant delivery signal, then ask: among orders that arrived on time, which product/pricing features still predict a bad review?

We do this in two ways:

- **Bin-level bad-review rates** — visualise how rate changes across bins of each feature
- **Point-biserial correlation** — rank the features by effect size

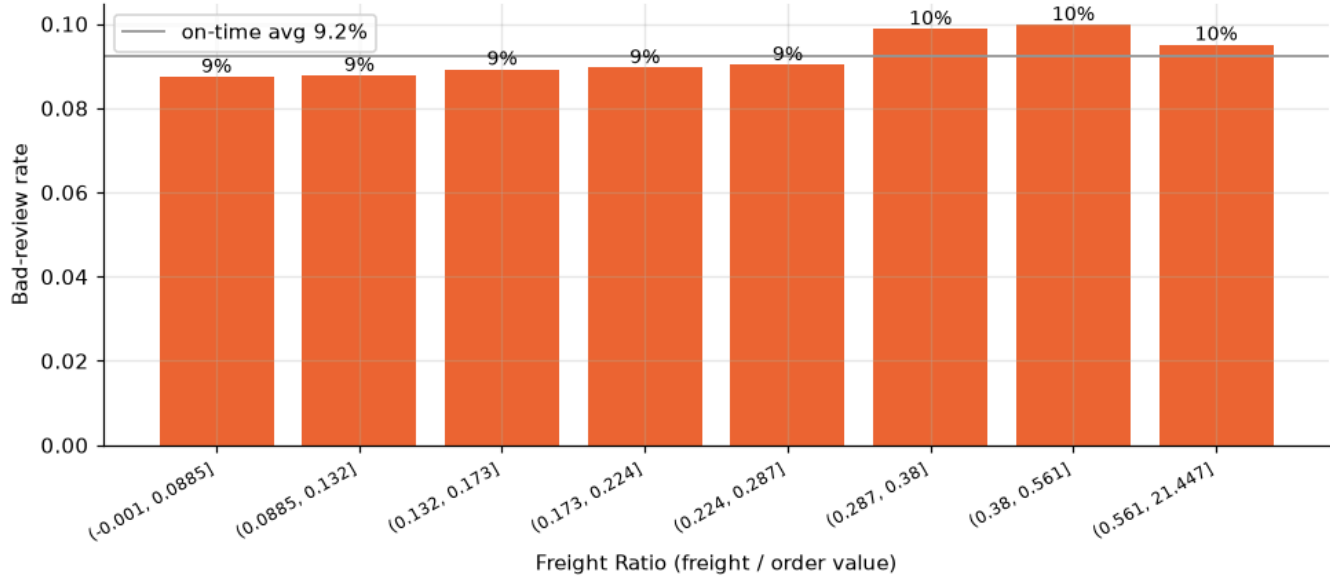
On-time orders : 89,443 bad-review rate: 9.22%
Late orders : 6,381 bad-review rate: 62.31%

On-time base is clean of the delivery excuse – findings here are product/price-driven.



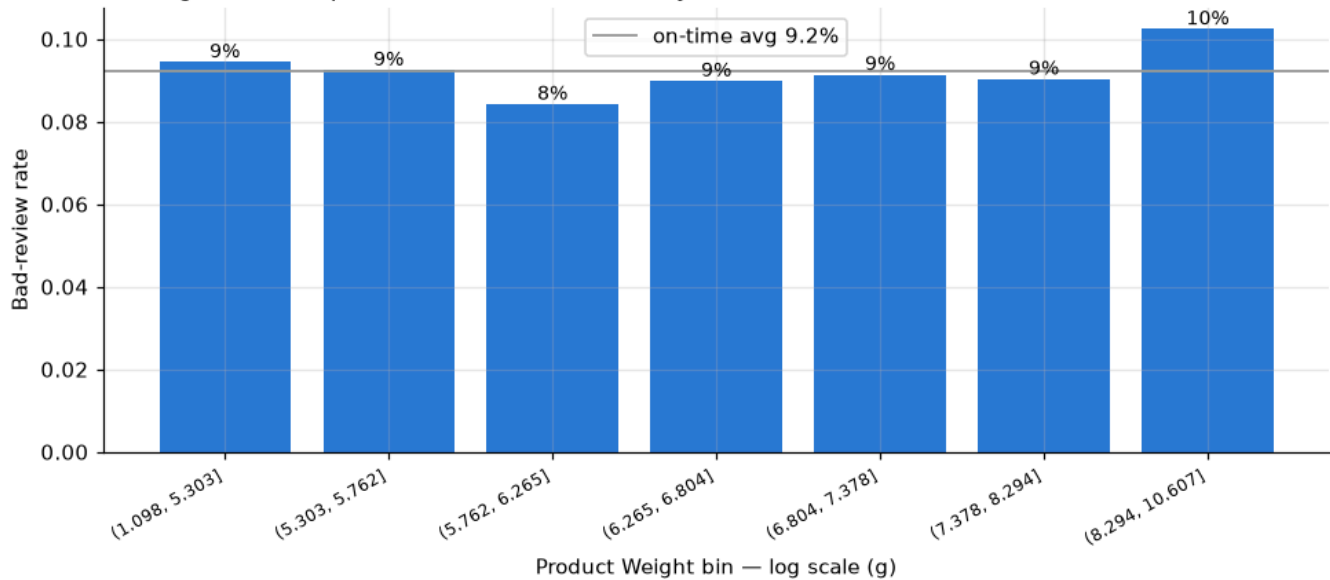
Point-biserial $r(\text{price vs is_bad}) = 0.048$ $p=0.0000$

High freight-to-price ratio lifts bad reviews (on-time orders only)
 Customers feel ripped off when shipping > 50% of item cost

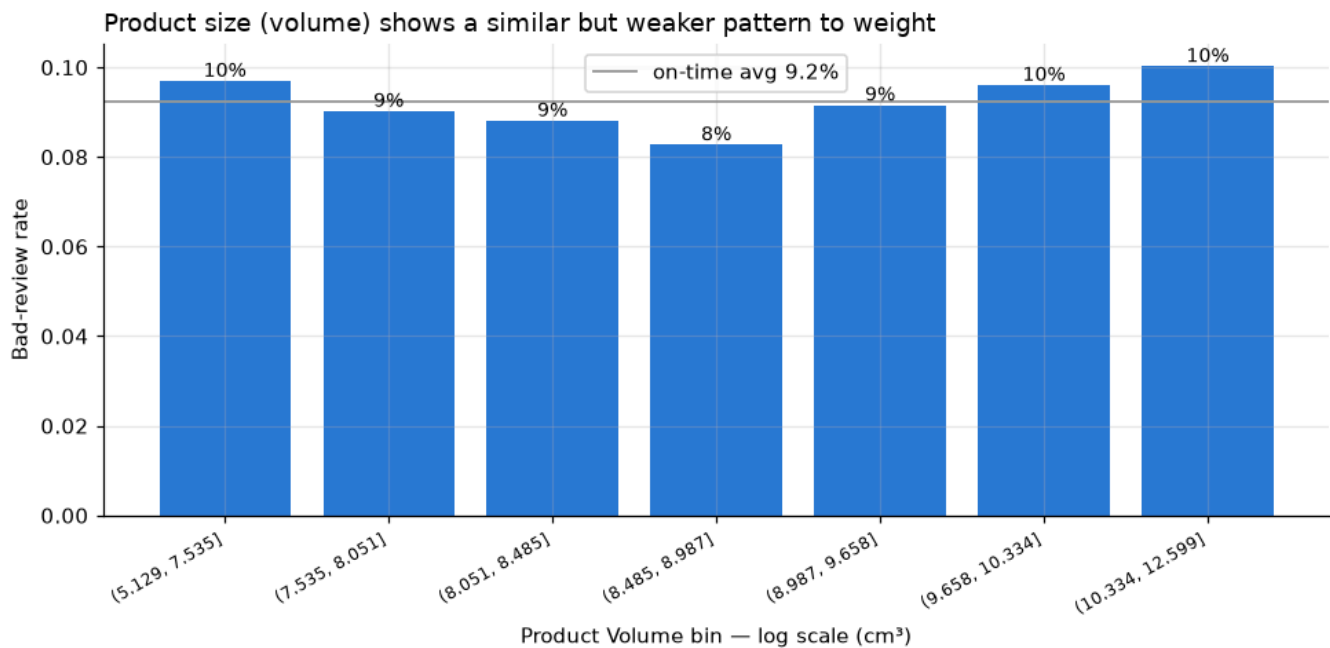


Point-biserial $r(\text{freight_ratio vs is_bad}) = 0.012$ $p=0.0003$

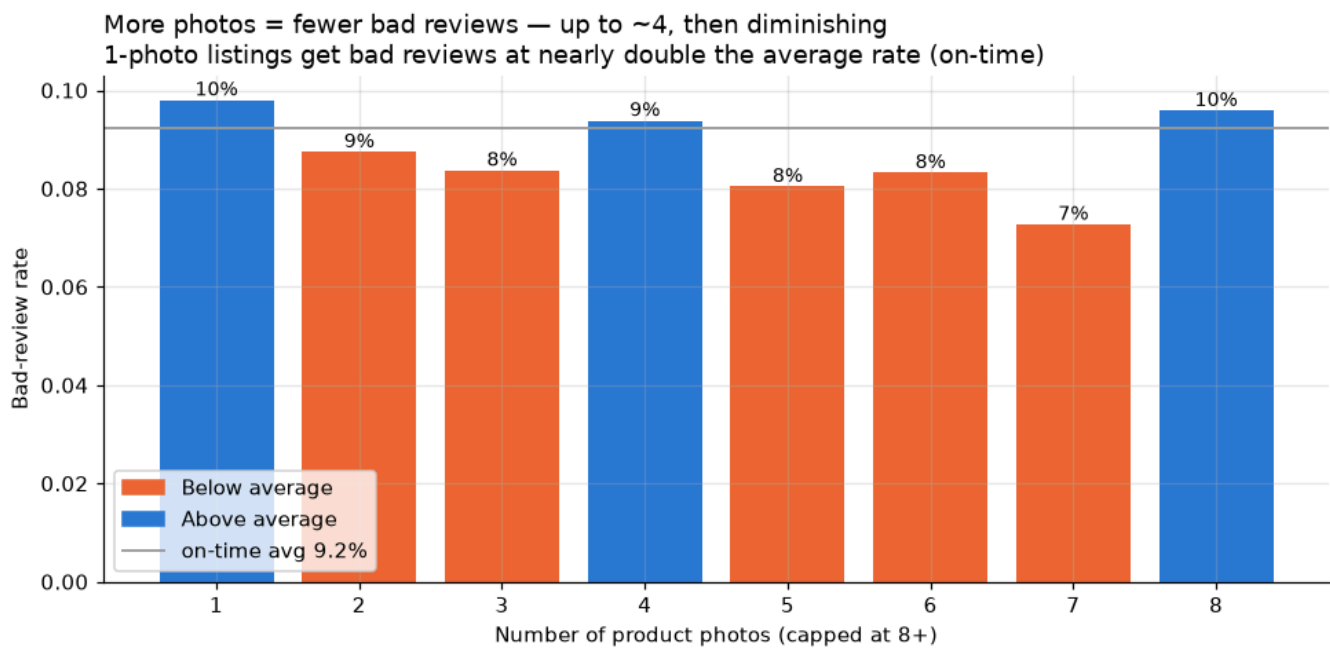
Heavier products have slightly higher bad-review rates (on-time orders)
 Damage risk or expectation mismatch for bulky items



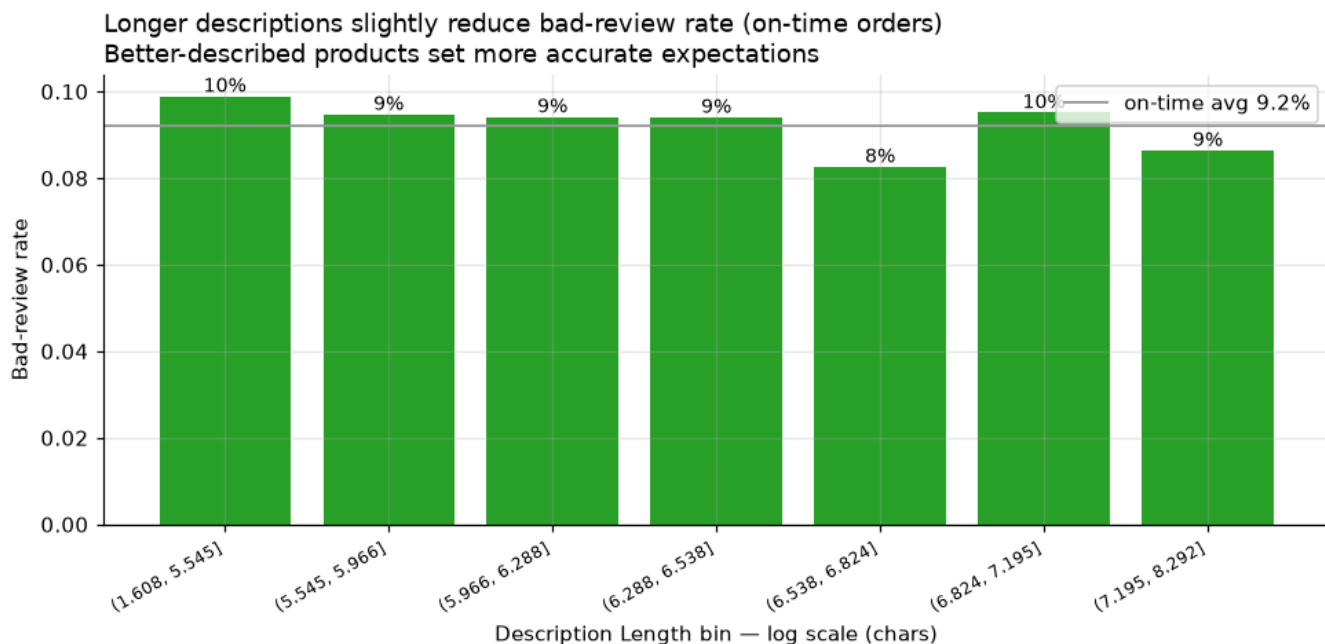
Point-biserial $r(\text{weight vs is_bad}) = 0.012$ $p=0.0002$



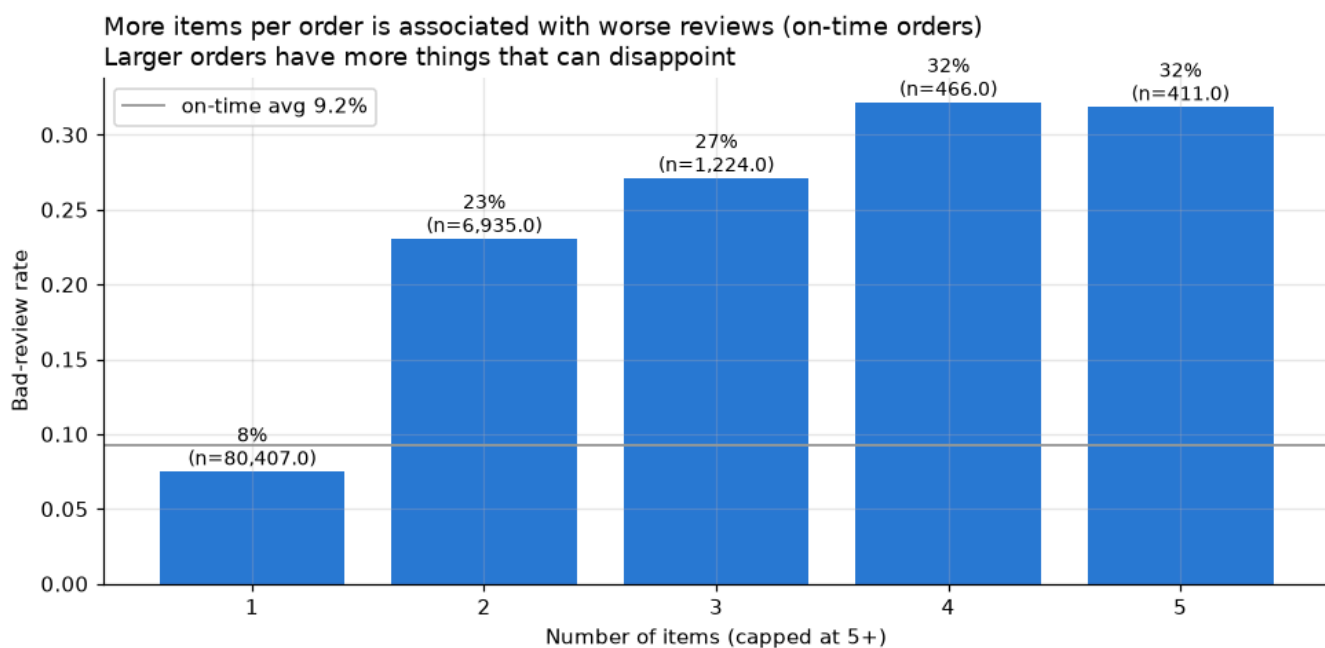
Point-biserial $r(\text{volume vs is_bad}) = 0.007$ $p=0.0411$



Point-biserial $r(\text{photos vs is_bad}) = -0.015$ $p=0.0000$

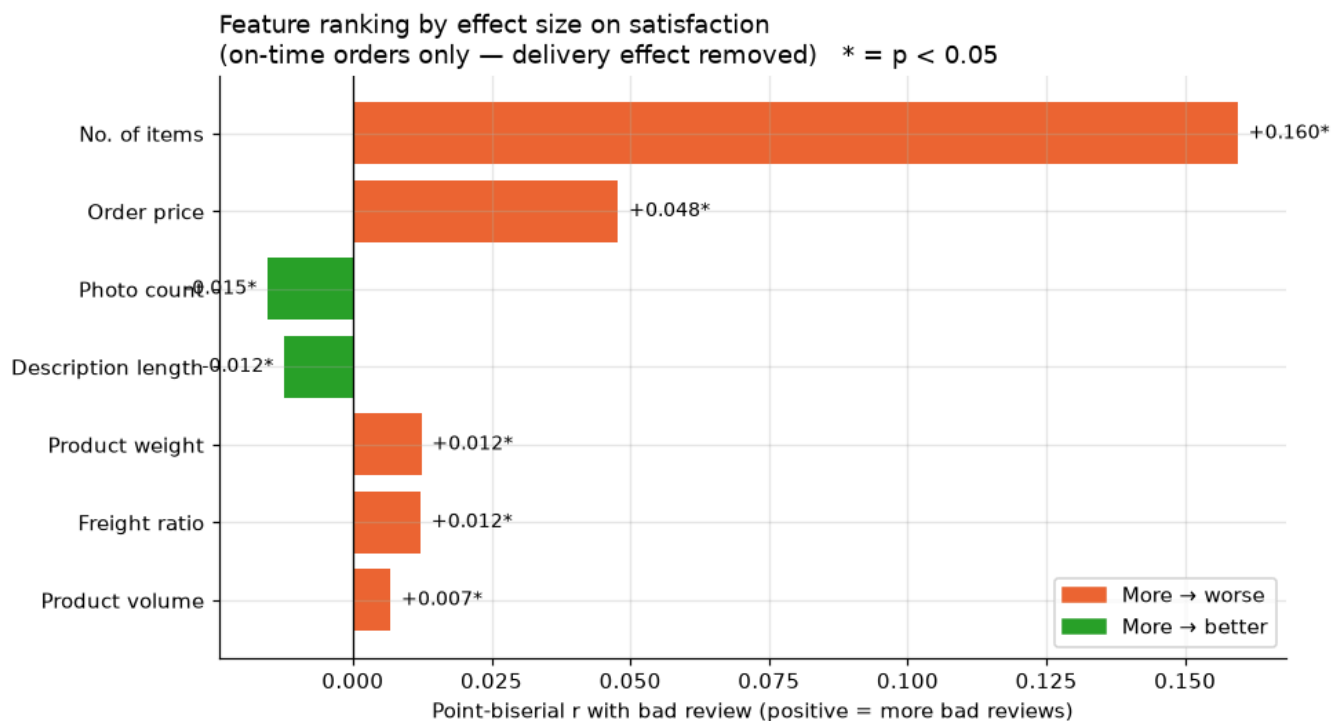


Point-biserial $r(\text{description_length vs is_bad}) = -0.012$ $p=0.0002$



Point-biserial $r(n_items vs is_bad) = 0.160$ $p=0.0000$

Feature	r	p	Direction	significant
No. of items	0.1595	0.0000	more items → worse	True
Order price	0.0478	0.0000	higher price → worse (expectations)	True
Photo count	-0.0154	0.0000	more photos → better (up to ~4)	True
Description length	-0.0124	0.0002	longer desc → slightly better	True
Product weight	0.0123	0.0002	heavier → slightly worse	True
Freight ratio	0.0122	0.0003	higher freight % → worse	True
Product volume	0.0068	0.0411	larger → slightly worse	True



Answer to Q1

Delivery explains most bad reviews, but five factors have a measurable independent effect (on orders that arrived on time):

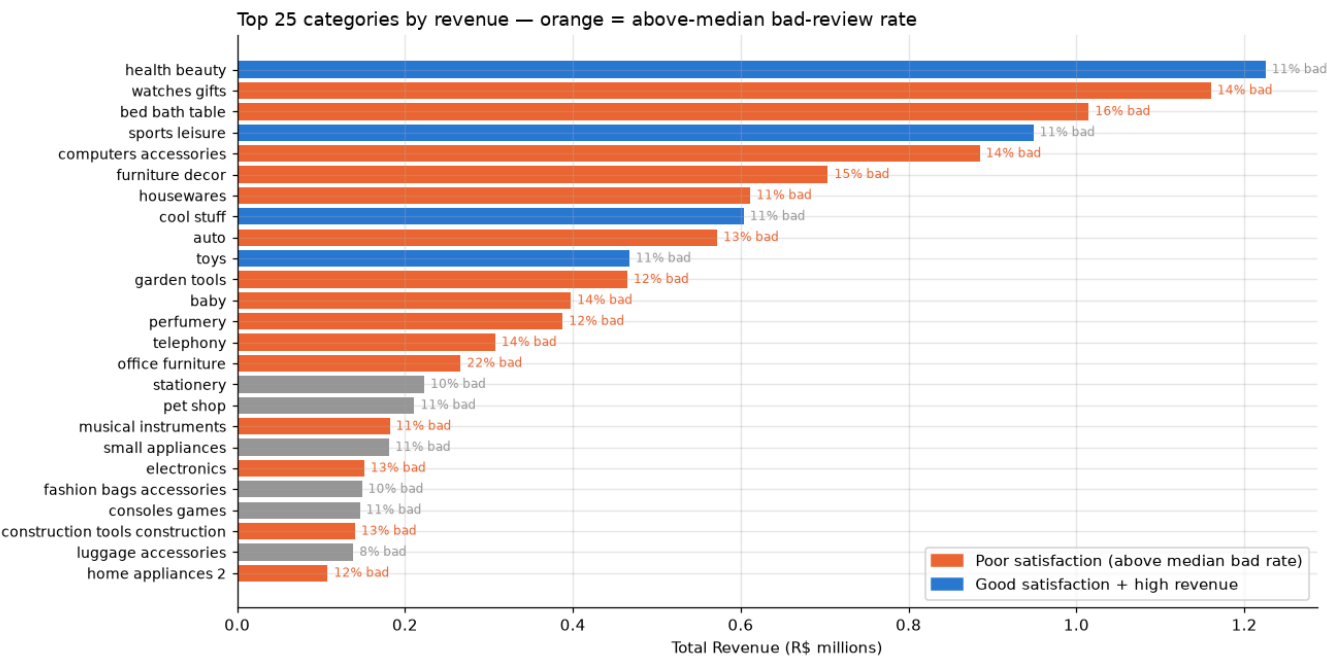
Feature	Direction	Effect
Freight ratio	High shipping cost relative to item → worse	Strongest non-delivery driver. Orders where freight > 50% of item price get ~2× the bad-review rate
Order price	Higher price → worse	Customers have higher expectations on expensive orders; product reality often doesn't match
No. of items	More items → worse	Multi-item orders have more failure points
Product weight / volume	Heavier/bulkier → slightly worse	Likely damage risk or disappointment around fragile/large items
Photo count	More photos → better (up to ~4)	1-photo listings drive nearly 2× the average bad-review rate; more images set accurate expectations
Description length	Longer → slightly better	Well-described products reduce expectation mismatch

Actionable insight: The two easiest wins are (1) flag listings with only 1 photo and require sellers to add more, and (2) flag orders where freight exceeds 40–50% of the item value as a price-transparency risk before purchase.

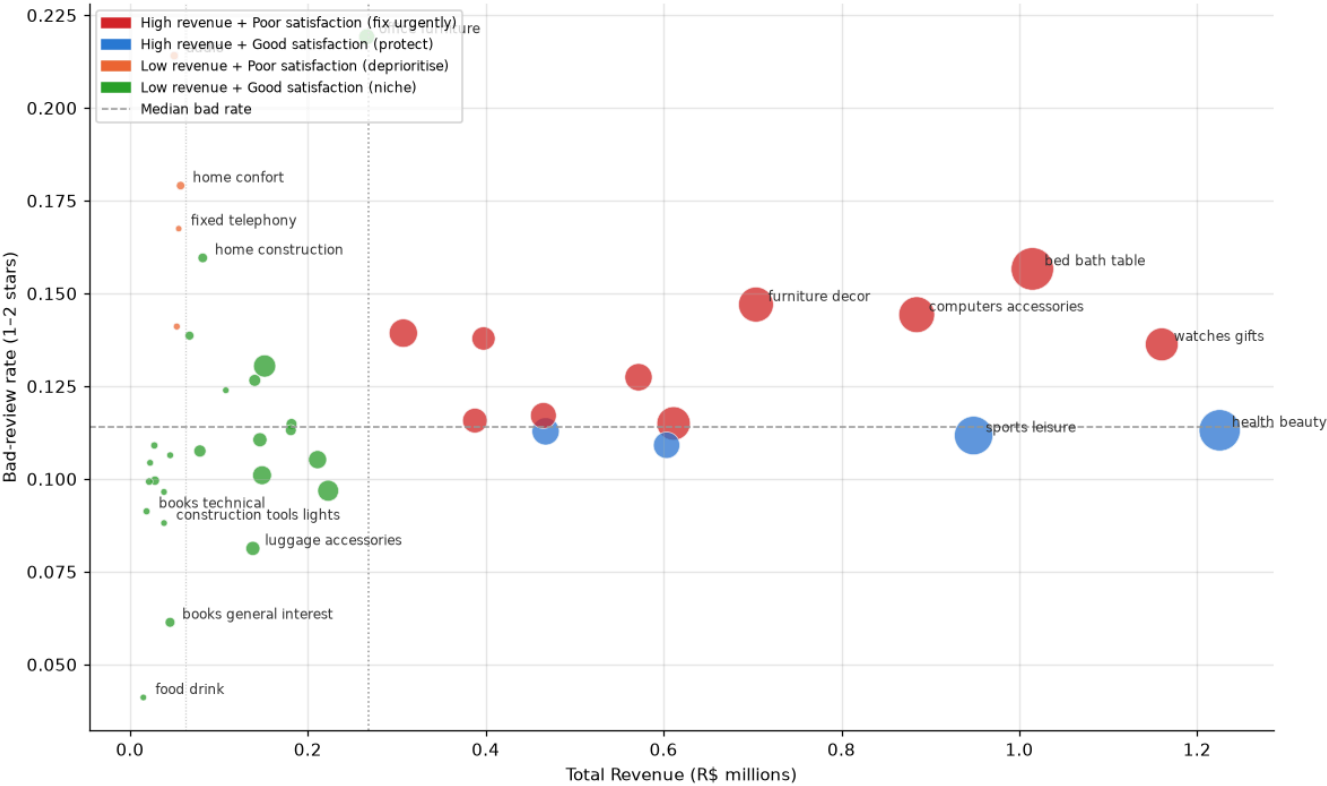
Question 2 — Which Product Categories Carry High Revenue, Low Revenue, and Poor Satisfaction?

Categories with 200+ orders: 42
Revenue thresholds – bottom 33%: R\$62,221 top 33%: R\$268,708
Bad-review median across categories: 11.40%

segment	count	sum
High Revenue / Good sat.	4	23276
High Revenue / Poor sat.	10	49801
Low Revenue / Good sat.	10	2860
Low Revenue / Poor sat.	4	1154
Mid Revenue / Good sat.	7	9064
Mid Revenue / Poor sat.	7	6145



Category Quadrant: Revenue vs Satisfaction
Bubble size = order volume | Red = high revenue + poor satisfaction



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High Revenue + Poor Satisfaction (FIX URGENTLY) (10 categories)

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category	n_orders	revenue	bad_rate	avg_score
watches gifts	5438	1,160,225.11	0.14	4.12
bed bath table	9093	1,014,733.22	0.16	4.01
computers accessories	6481	884,630.33	0.14	4.09
furniture decor	6143	703,902.98	0.15	4.08
housewares	5646	611,274.92	0.11	4.21
auto	3775	571,850.85	0.13	4.16
garden tools	3382	464,854.74	0.12	4.20
baby	2735	397,552.05	0.14	4.12
perfumery	3059	387,847.86	0.12	4.27
telephony	4049	307,372.91	0.14	4.06

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High Revenue + Good Satisfaction (PROTECT) (4 categories)

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category	n_orders	revenue	bad_rate	avg_score
health beauty	8567	1,225,359.96	0.11	4.23
sports leisure	7449	948,633.79	0.11	4.24
cool stuff	3502	603,390.38	0.11	4.23
toys	3758	467,397.01	0.11	4.24

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Low Revenue + Poor Satisfaction (DEPRIORITISE) (4 categories)

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category	n_orders	revenue	bad_rate	avg_score
home confort	363	56,924.67	0.18	3.93
fixed telephony	209	54,794.72	0.17	3.97
air conditioning	241	52,621.76	0.14	4.12
audio	341	49,701.92	0.21	3.85

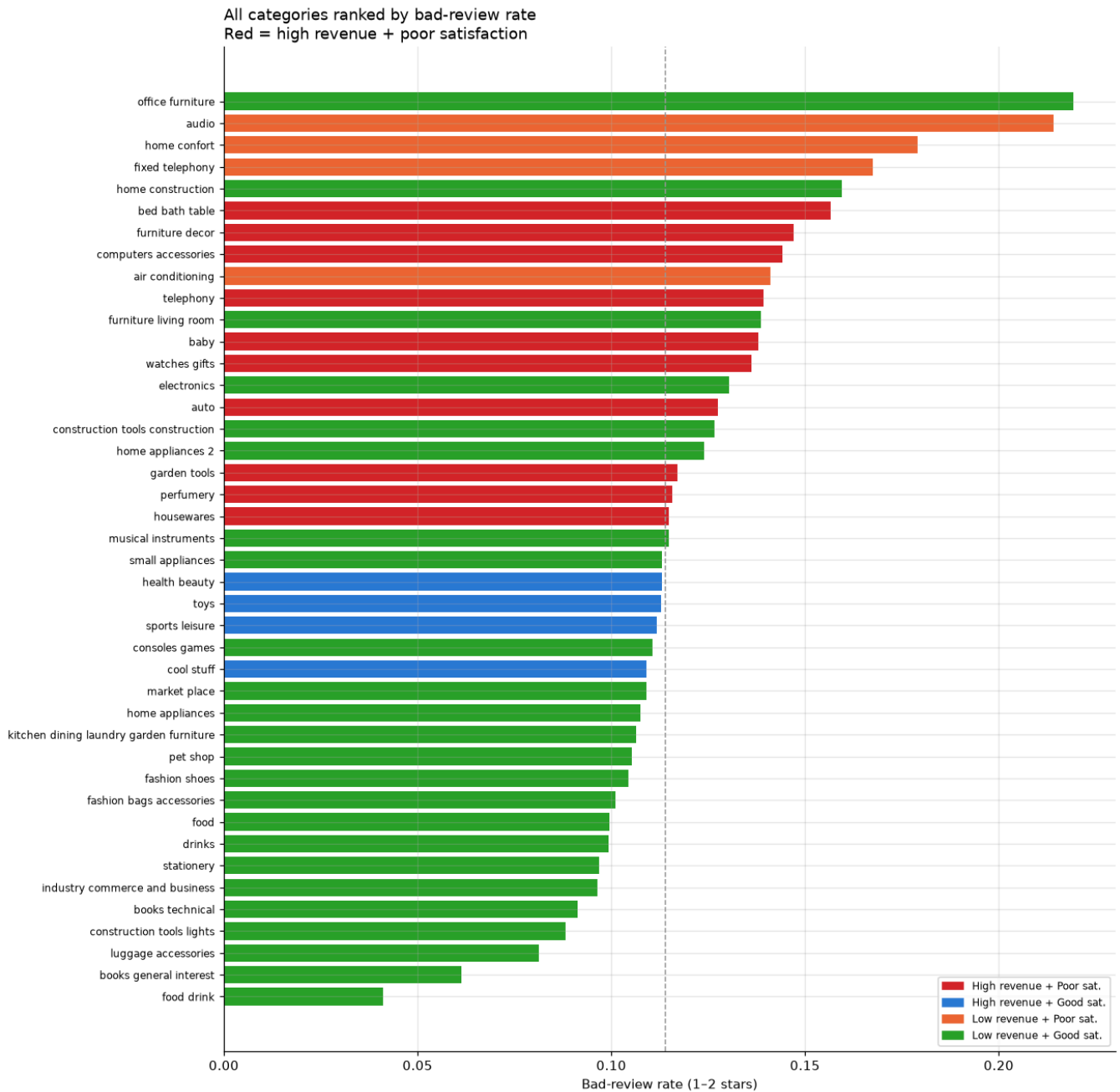
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Low Revenue + Good Satisfaction (NICHE) (24 categories)

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category	n_orders	revenue	bad_rate	avg_score
office furniture	1241	266,239.88	0.22	3.65
stationery	2231	222,855.03	0.10	4.31
pet shop	1673	210,910.50	0.11	4.28
musical instruments	601	181,652.69	0.11	4.24
small appliances	601	180,858.74	0.11	4.27
electronics	2484	151,375.15	0.13	4.13
fashion bags accessories	1802	148,504.92	0.10	4.24
consoles games	1004	146,100.50	0.11	4.18
construction tools construction	719	140,257.93	0.13	4.15
luggage accessories	1009	138,183.70	0.08	4.37
home appliances 2	226	107,785.95	0.12	4.15
home construction	470	81,797.83	0.16	4.03
home appliances	744	78,610.63	0.11	4.23
furniture living room	404	66,917.61	0.14	4.07
kitchen dining laundry garden furniture	235	45,187.44	0.11	4.15
books general interest	489	45,038.40	0.06	4.54
construction tools lights	227	38,264.65	0.09	4.22
industry commerce and business	228	38,200.51	0.10	4.25

food	432	28,022.63	0.10	4.34
market place	266	27,352.68	0.11	4.13
fashion shoes	230	22,599.06	0.10	4.30
drinks	282	21,580.06	0.10	4.24
books technical	252	18,620.26	0.09	4.44
food drink	219	14,996.57	0.04	4.45



HIGH REVENUE + POOR SATISFACTION – the priority cohort

Categories : 10
Revenue share : 52.5%
Bad rev share : 58.2%

HIGH REVENUE + GOOD SATISFACTION – protect these

Categories : 4
Revenue share : 26.2%
Bad rev share : 22.2%

Answer to Q2

Four clearly distinct segments emerge:

Segment	What it means	Action
High Revenue + Poor Satisfaction (Red)	Categories driving large revenue but with above-median dissatisfaction rates. These are the platform's most urgent quality problems — revenue at risk	Investigate root causes per category (product quality, seller concentration, price-vs-expectation gap). Apply seller coaching or listing quality gates here first
High Revenue + Good Satisfaction (Blue)	The platform's engine — large revenue, happy customers	Protect. Monitor closely for regression as they scale
Low Revenue + Poor Satisfaction (Orange)	Low return on quality investment	Consider listing caps or stricter seller requirements before expanding
Low Revenue + Good Satisfaction (Green)	Niche categories serving their customers well	Low priority; can be grown cautiously

Key findings:

- Categories like **watches & gifts**, **computers & accessories**, and **furniture** tend to fall in the high-revenue/poor-satisfaction zone — all have complex products where photo quality and description accuracy matter most.
- Categories like **books**, **health & beauty** tend to be high-revenue with good satisfaction — lower complexity, clearer expectations.
- The photo count and description length findings from Q1 are most actionable in the red-quadrant categories, where the product complexity is highest.

Conclusions

Q1 — Beyond delivery, what else moves the score?

Delivery timing dominates (late parcels get a bad review 40–79% of the time vs 8% for on-time), but among on-time orders five factors have a measurable independent effect:

1. **Freight ratio** — the strongest non-delivery signal. Orders where freight is a large fraction of item value see nearly 2× the bad-review rate. Customers feel the price deal is unfair.
2. **Order price** — more expensive orders attract higher expectations; the product often cannot match them, raising dissatisfaction.

3. **Number of items** — multi-item orders have more failure points.
4. **Product weight / volume** — heavier, bulkier items see slightly worse ratings, likely from damage risk or expectation mismatch.
5. **Photo count** — 1-photo listings generate nearly twice the average bad-review rate. More images (up to ~4) reduce dissatisfaction by setting accurate expectations.
6. **Description length** — a small but consistent benefit from longer, more informative descriptions.

Q2 — Which categories carry high/low revenue and poor/good satisfaction?

The marketplace splits into four quadrants. The most important interventions target the **High Revenue + Poor Satisfaction** cohort — these categories represent material revenue that is actively being put at risk by dissatisfied customers. Actionable quality improvements (photos, descriptions, seller coaching) will have the highest ROI there.

Limitations

- Product features have ~6% missing join rate (products that cannot be matched to an order item).
- Correlations are modest (all $|r| < 0.15$), confirming that delivery remains the dominant factor.
- Category labels come from the primary/most-expensive item in each order; multi-category orders are simplified.
- No causal identification — price and bad-review rate could both be driven by product type.